

updates increased the speed and range of the cars.

In the wearables space, Jawbone launched a software update last year that effectively doubled the devices' battery life. From a user standpoint, this was very satisfying, as it meant that the user only had to charge the device twice a month as opposed to once a week, according to Arvind Sanjeev, founder of *DIYHacking.com* and creator of the indigenous Google Glass clone.

Sanjeev adds that the next thing would be the charging port of the device. Since conventional micro-USB slots make the device bulky, custom solutions and docking ports should be designed specifically for each wearable.

Better data-management process

"The most interesting challenge while designing our wearable was to figure out how to collect the data we need and still use extremely low power. Our device, a pair of smart insoles that monitor a runner's running style and form while he or she is running, uses a small rechargeable battery. We had to invent a novel way of collecting data that could get us all the data we need but consume extremely little power," explains Anshuman Singh, co-founder of ReTiSense and inventor of Strida-lyser.

The primary design challenges that go into these wearables start with the selection of the right low-power sensors and low-power communication technology. Optimising the functioning of the device to minimise the current drawn by the product and its sensors, and power profile for communication of data to the smartphone or other companion device, could help control power consumption.

"The advice I would give is to use the basic principles of a good design; collect only the data that you really need and only with the frequency you need it. Make sure to do as little processing as possible and communicate data as late and as infrequently as possible," Singh adds.

Data processing needs to be moved to the gateway; in the case of most

TABLE I
Display and Battery Technology Compared

Product	Processor	Battery	Display
Fitbit Flex	• STM32L151C6 • STMicroelectronics • ARM Cortex-M3 32-bit MCU, 128kB flash, 16kB SRAM, 4kB EEPROM, 12-bit ADC, 12-bit DAC, LCD controller • Interfaces: USB 2.0 USART, SPI and I2C	• One lithium-polymer battery; rechargeable • Lasts five days	• Five white LED indicator lights
Garmin Vivofit	• nRF51422 • Nordic Semiconductor • Multi-protocol (ANT, ANT+, Bluetooth Smart etc) SoC • Integrates 2.4GHz transceiver, 32-bit ARM Cortex-M0 CPU, 256kB flash, 16kB RAM and analogue and digital peripherals	• Two Panasonic CR1632 3V lithium-coin batteries; replaceable • Lasts one+ years	• 2.9cm (1.15-inch), 5-segment LCD
Microsoft Band	• Kinetis K24 • Freescale • 32-bit ARM Cortex-M4, USB 2.0 on-the-go 1MB flash with 4-level security protection 256kB SRAM, two 16-bit ADCs and two 12-bit DACs	• Two 100mAh lithium-polymer batteries; rechargeable • Typical: Two to three days • Lasts up to three days	• Capacitive 3.6cm (1.4-inch) TFT full-colour (320 x 106 pixels)
Samsung Gear Fit	• STM32F439ZL • STMicroelectronics • DSP with FPU, ARM Cortex-M4 MCU with 2MB flash, 180MHz CPU, art accelerator, hardware crypto, SDRAM and TFT	• One 210mAh battery; rechargeable • Typical: Two to three days • Lasts up to five days	• 4.7cm (1.84-inch) curved super-AMOLED touchscreen display (432 x 128 pixels)

Use the basic principles of a good design; collect only the data that you really need and only at the frequency you really need it

wearables, this gateway is the smartphone. This is where the right hardware-software co-design aspect becomes critical.

Wearable devices can include many combinations of biometric and physiological sensors, depending upon the data that these are designed to collect. These data can include distance walked, calories burned, heart rate, temperature, blood-oxygen level, sleep quality, pressure and others.

"Specifics of what is being measured drive the selection of MCUs, signal-conditioning components and connectivity chips that are used in a wearable device. Some key requirements for a wearable device are low power, connectivity and sensor-data

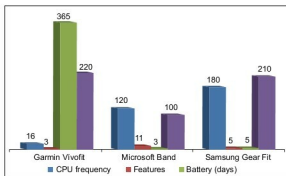


Chart comparing clock speed, features and battery life

processing capability," explains Smith.

Wireless connectivity allows wearable devices to interface with smartphones, tablets, PCs and other intelligent devices. Agency-certified, low-power Bluetooth, Bluetooth Smart and Wi-Fi modules can satisfy the requirement to get to market quickly, and can save significant time in RF design and regulatory approvals.